

Khawaja Yunus Ali University

School of Science and Engineering

Dept. of Computer Science and Engineering

Mid-term Examination: Summer 2018 Semester

Program: B.Sc. in CSE (Batch: 6th) 1st Year 1st Semester

Course Code: PHY 111

Course Title: Physics – I

Time: 1 hour & 30 minutes

Total Marks: 30

Group-A

[Answer any 5 (Five) Questions]

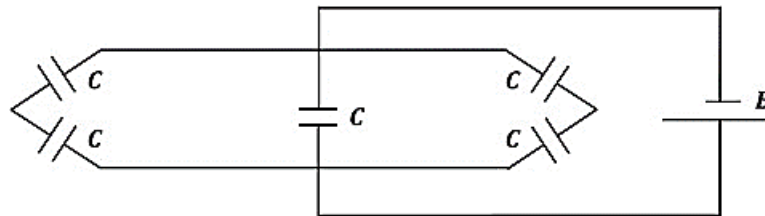
5 × 5 = 25

1. Read the stem given below and answer the question that follows:

“Two charges $q_1 = 1.0 \times 10^{-6}$ coul and $q_2 = 2.0 \times 10^{-6}$ coul are kept 10cm away from each other.”

- a) Calculate the repulsive force exist between two charges? 2
- b) At what point the line joining the two charges is the electric field strength zero? 3

2. See the curve given below as stem and answer the question that follows:



Combination of capacitors of capacitance, $C = 10\mu F$

- a) Define 1 Farad. 1
 - b) Calculate the equivalent capacitance of above mentioned circuit diagram. 3
 - c) Mention a combination of given capacitors to get the minimum equivalent capacitance. Give answer with diagram. 1
3. a) Give some uses of capacitor. 1
- b) Is it possible to charge a capacitor to a very high potential? 1
- c) The parallel plate of a Ruby-Mica filled capacitor are every 2.0mm apart. 3
What must be the plate area be if the capacitance is to be 2.0 F? Given dielectric constant for Ruby-Mica is 1.5.

4. a) What is current density? 1
- b) A silver wire 1 mm in diameter carries a charge of 90 coulomb in 1 hr. Silver contains 5.8×10^{28} free electrons/m³. Calculate (i) the current in the wire (ii) current density and drift velocity of electrons in the wire. 4
5. Read the stem given below and answer the following question:
- “By applying Gauss Law, we can drive an equation for electric field due to a long straight charged wire”*
- b) Justify the above statement. 3
- c) Is it possible to drive coulomb’s law from above mentioned law?-Justify your answer. 2
6. a) Write down a short note on “Quantization of Electric Charge” 2
- b) Define: (any three) 3
- i) Gaussian surface
 - ii) Dielectric constant
 - iii) Electric Lines of Force
 - iv) Electrostatics
 - v) Equipotential Surface.
7. a) Distinguish electric potential and electric potential energy. 2
- b) $q_1 = 2 \times 10^{-8}c$, $q_2 = -2 \times 10^{-8}$, $q_3 = 3 \times 10^{-8}c$ and $q_4 = -3 \times 10^{-8}c$ charges are situated at the corners of a square shape. If the length of one arm of the square shape is 1cm, what will be the electric potential at center of the shape? 3

[Write the appropriate answer of all questions in the answer script sequentially]

1. Minimum amount of charge in universe is-

- | | | | |
|----|-------------------------|----|-------------------------|
| a. | $1.6 \times 10^{-16} C$ | b. | $1.9 \times 10^{-16} C$ |
| c. | $1.9 \times 10^{-19} C$ | d. | $1.6 \times 10^{-19} C$ |

2. Potential energy of a capacitor –

- i. $W = \frac{1}{2} QV$
 ii. $W = \frac{1}{2} CV^2$
 iii. $W = \frac{1}{2} \frac{Q^2}{C}$

Which one is right?

- | | | | |
|----|----------------------|----|-------------------------|
| a. | $i \text{ \& } ii$ | b. | $i \text{ \& } iii$ |
| c. | $ii \text{ \& } iii$ | d. | $i, ii \text{ \& } iii$ |

3. With an equality or balance of charge, the object said to be-

- | | | | |
|----|----------------------|----|--------------------|
| a. | electrically neutral | b. | negatively charged |
| c. | positively charged | d. | maximum charged |

4. What is the value of $\frac{1}{4\pi\epsilon_0}$?

- | | | | |
|----|--|----|-------------------------------|
| a. | $1.6 \times 10^{-16} Nm^{-1}$ | b. | $9 \times 10^9 Nm^2 C^{-2}$ |
| c. | $8.85 \times 10^{-12} C^2 N^{-1} m^{-2}$ | d. | $6.6 \times 10^{-16} Nm^{-1}$ |

5. The potential differences between two points in an equipotential surface is -

- | | | | |
|----|---|----|---|
| a. | 0 | b. | 1 |
| c. | 2 | d. | 4 |

6. What is the Relation between electric field and electric force?

- | | | | |
|----|---------------------|----|---------------------|
| a. | $E = \frac{q_0}{W}$ | b. | $E = \frac{F}{q_0}$ |
|----|---------------------|----|---------------------|

c. $E = \frac{F}{W}$

d. $V = \frac{W}{E}$

7. What is the unit of the electric permittivity?

a. *Volt*

b. *Joule*

c. *None*

d. *E*

8. The net flux of a cylindrical Gaussian surface placed in an external field is

a. negative

b. positive

c. infinity

d. Zero

9. Which one is possible?

a. $q = 45e$

b. $q = 35.1e$

c. $q = 25.5e$

d. $q = 15.9e$

10. Conductance is to the cross-sectional area of the conductor.

a. equal

b. proportional

c. inversely proportional

d. independent

-----The End-----